

Lecture 19: Discrete-Time Markov Chains

Jamie Haddock

Table of contents

- 1 Markov chains::: **{.callout-note icon=false}## Definition: Markov property**
 $X_0, X_1, \dots$ has the Markov property if

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = P(X_{n+1} = j \mid X_n = i) = p_{ij}$$

– the future depends on the past only through the present.:. . .::: **{.callout-warning icon=false}## Fact**The p_{ij} form the transition matrix P ; n -step transition probabilities are given by P^n .:—**## A simple weather chain** 1

1.1 Reading & HomeworkReading: §7.1-7.2 (Bertsekas & Tsitsiklis, *Introduction to Probability*). . 1

—title: “Lecture 19: Discrete-Time Markov Chains”author: “Jamie Haddock”format: revealjs: output-file: Lecture19_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

- 1 Markov chains::: **{.callout-note icon=false}## Definition:**
Markov propertyA process $X_0, X_1, \dots$ has the Markov property if

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = P(X_{n+1} = j \mid X_n = i) = p_{ij}$$

– the future depends on the past only through the present.:. . .::: **{.callout-warning icon=false}## Fact**The p_{ij} form the transition matrix P ; n -step transition probabilities are given by P^n .:—**## A simple weather chain**

```
P = np.array([[0.8, 0.2], [0.4, 0.6]])x = 0history = [x]for _ in range(2000):    x = rng.choice(2, p=P[
```

—# Case Study: Randomized Kaczmarz as a Markov Chain x_{k+1} is built from x_k and a *fresh, independent* random row draw – it does not depend on $x_{k-1}, x_{k-2}, \dots$ once x_k is known. That’s the Markov property, on the continuous state space $\mathbb{R}^n$.

```
m, n = 20, 5A = rng.normal(size=(m, n))x_true = rng.normal(size=n)b = A @ x_truep = np.sum(A**2, axis=1,
```

- 1.1 Reading & HomeworkReading: §7.1-7.2 (Bertsekas & Tsitsiklis, *Introduction to Probability*).